



ANM — Research Mode (Full Blueprint)

1. What Research Mode Is

Research Mode is ANM's highest-intelligence operating mode. It is explicitly designed for **deep research, scientific reasoning, technical analysis, simulations, and documentation**.

Unlike FAST or AUTO modes, Research Mode: - prioritizes correctness over speed - enforces authority models - runs modules in parallel - documents results formally

It is **opt-in only** and never enabled by default.

2. Core Philosophy

Correctness > Authority > Completeness > Speed

Research Mode is willing to: - wait longer - consume more compute - return uncertainty

if that leads to a more accurate and honest result.

No silent fallbacks. No guessing.

3. Authority Model Assignment (Locked)

Each specialist module uses exactly **one LLM**. No overlap, no voting.

Science & Mathematics Authority

- Physics
- Chemistry
- Biology
- Mathematics

LLM: NanBeige4-3B

Role: - conceptual scientific reasoning - equations and derivations - academic-style explanations

Meta-Cognition & Self-Awareness

- self-audit
- reasoning validation
- uncertainty estimation

LLM: DeepSeek R1-1.5B

Role: - check reasoning quality - detect inconsistencies - assess confidence

Internet Research & Structured Output

- web reading
- summarization
- information extraction
- structured outputs (JSON, tables)

LLM: Qwen2.5-3B-Instruct

Role: - read noisy internet data - summarize sources - prepare research inputs

Coding Authority

- programming
- algorithms
- system design

LLM: StableCode-3B

Role: - generate correct code - explain implementations - validate technical logic

4. Router Behavior

Research Mode uses a **deterministic router**.

Router rules: - strict domain detection - hard domain → model binding - no probabilistic voting - no early exits - no fast fallback

If a domain is detected, its authority model **must run**.

5. Parallelism Model

Research Mode uses **maximum parallelism**.

- Parallelism scale: $X / 10$
- X = number of active modules

If 8 modules are relevant, **all 8 run in parallel**.

Execution graph: - Directed Acyclic Graph (DAG) - Independent modules run concurrently - Synchronization only at aggregation points

6. Web-of-Thought (WoT)

WoT is **mandatory** in Research Mode.

Rules: - minimum reasoning depth enforced - no "good enough" termination - full aggregation always executed - DeepSeek R1 is the reasoning spine

WoT coordinates: - specialist outputs - intermediate reasoning steps - final synthesis

7. Internet Research Pipeline

When external knowledge is required:

1. Fetch sources
2. Read and summarize (Qwen)
3. Reason and connect (R1)
4. Validate:
5. science/math → NanBeige
6. code → StableCode
7. Final synthesis (R1)

Independent steps run in parallel.

8. Memory Usage

Current Implementation

Diary Memory - always read before reasoning - always written after completion - stores reasoning trails, attempts, failures - treated as context, not factual authority

Planned (Future, Not Faked)

Vector Database (VB) - validated long-term knowledge - semantic retrieval

RAG (Retrieval-Augmented Generation) - internal + external grounding - feeds research pipeline

Research Mode is designed to fully exploit VB and RAG once implemented.

9. Failure Policy

Research Mode enforces **scientific honesty**.

- authority failure is explicitly reported
- retries allowed

- if still failing → uncertainty returned

No silent downgrades.

10. Final Output: Research PDF

Every Research Mode run ends with a **structured PDF**.

PDF Contents

1. Title Page
2. Executive Summary
3. Research Question & Scope
4. Sources & Data
5. Analysis (Science / Code / Logic)
6. Meta-Cognition & Cross-Checks
7. Limitations & Uncertainty
8. Final Conclusions
9. Appendix (models used, WoT depth, logs)

The PDF is generated from **validated internal outputs**, not raw chat text.

11. What Research Mode Is NOT

- not casual chat
- not fast-response mode
- not demo-optimized

It **is**: - research-grade - authority-driven - parallel - transparent

12. Summary

Research Mode is ANM operating at **full potential**.

FAST Mode answers quickly. AUTO Mode survives safely.

Research Mode thinks deeply, checks itself, and documents the result.

This blueprint defines ANM as a true research system, not a chatbot.